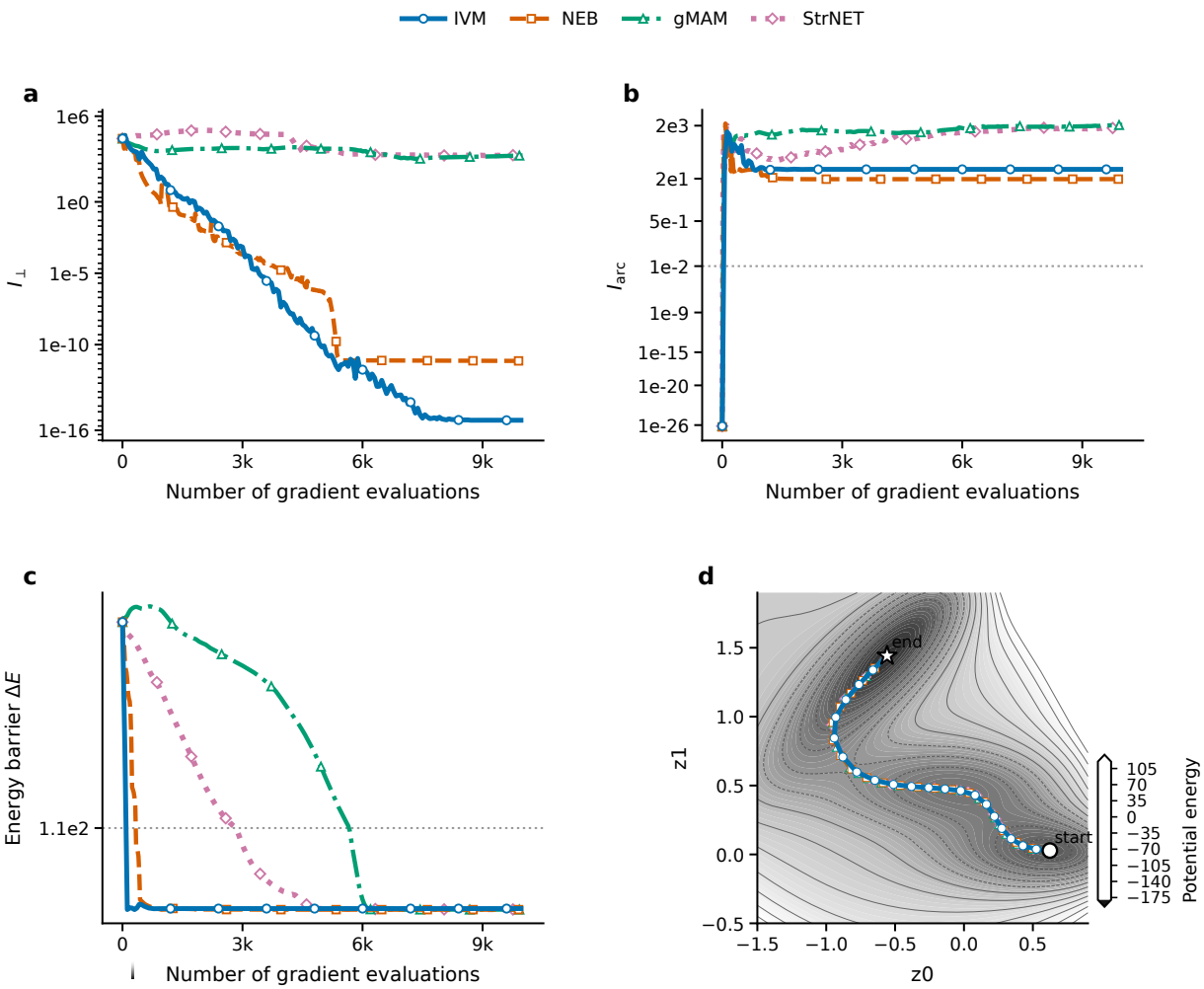

IVM
NEB
gMAM
StrNET
a
1e6
1e0
1e-5
1e-10
1e-16
$I_\perp$
0
3k
6k
9k
Number of gradient evaluations
b
2e3
2e1
5e-1
1e-2
1e-9
1e-15
1e-20
1e-26
I_{arc}
0
3k
6k
9k
Number of gradient evaluations
c
Energy barrier ΔE
1.1e2
0
3k
6k
9k
Number of gradient evaluations
d
end
start
z1
1.5
1.0
0.5
0.0
−0.5
−1.5
−1.0
−0.5
0.0
0.5
z0
105
70
35
0
−35
−70
−105
−140
−175
Potential energy